

The Operating Boundary of Reliability-Aware Multimodal Fusion: A Provable Impossibility and Its Complement

Abstract

Reliability-aware multimodal fusion—down-weighting a sensor whose score distribution drifts—is widely assumed to be a general improvement over simple averaging. We show it is not, and we prove exactly where its value lies. Our central result is a *negative* one: on clean, near-separable transfer, no fusion rule (any reliability gate included) can beat the confidence-weighted mean by a statistically certifiable margin, because the mean already attains the Neyman–Pearson ceiling. We turn this into a *computable impossibility certificate* with an oracle-credibility guard that prevents the impossibility from being manufactured by an under-powered estimator. Complementary, we prove the matching positive: under differential-reliability stress—coherent or k -of- D modality corruption—every reliability-blind aggregator (including the median) is dominated, so a reliability gate has a provable lower-bounded advantage. Together these results give a sharp *operating boundary*: reliability gating is neither universal nor useless; it helps precisely when modalities have differential reliability and provably cannot help when they do not. We further contribute a validation-only degenerate-channel guard that makes even the confidence-weighted mean robust to sign-inverted or saturated experts, and we illustrate the boundary empirically on industrial multimodal anomaly benchmarks. This reframes reliability-aware fusion from a method one tunes to a mechanism one deploys *within a characterized envelope*.

1 Introduction

Multimodal anomaly detection fuses per-modality (per-sensor) scores into a single decision. A recurring intuition is that a *reliability gate*—a component that detects when a modality’s scores drift from a validated reference and down-weights it—should generally outperform a fixed aggregator such as the confidence-weighted mean (CW). The empirical record is mixed: reliability gates sometimes win by large margins and sometimes tie or lose, and the field lacks a principled account of *when* each occurs.

We give that account, and it is sharp. Rather than proposing yet another fusion architecture and reporting where it happens to win, we ask the structural question—*when can reliability gating help at all?*—and answer it with two matching theorems and a computable certificate. The contribution is a law of the problem, not a method tuned to a benchmark:

- **An impossibility (Section 3).** On clean, near-ceiling transfer the confidence-weighted mean already reaches the Neyman–Pearson ranking ceiling, so the recoverable advantage of *every* fusion class is below the minimum detectable effect. We certify this with an oracle-guarded test that cannot be gamed by a weak estimator.
- **The matching positive (Section 4).** Under coherent or sparse k -of- D modality corruption, every reliability-blind aggregator is dominated by a reliability-aware oracle with a closed-form lower bound on the gap.

- **The boundary (Section 5) and a robustness guard (Section 6).** The two results meet at a characterizable boundary; a validation-only degenerate-channel guard removes a failure mode that otherwise corrupts *any* aggregator.

2 Setup

Let $Y \in \{0, 1\}$ be an anomaly label and $S = (S_1, \dots, S_M)$ a vector of per-modality scores with class-conditional laws. For a fusion rule $g : \mathbb{R}^M \rightarrow \mathbb{R}$, write its ranking quality as the AUROC $A(g) = \Pr[g(S^+) > g(S^-)]$, with $S^+ \sim (S \mid Y=1)$ and $S^- \sim (S \mid Y=0)$ independent. The confidence-weighted mean is the linear rule $g_{\text{CW}}(s) = \langle w, s \rangle$. A *reliability gate* is any rule in a class $\mathcal{G}$ that may re-weight modalities using a validation-derived reliability signal; $\mathcal{G}$ contains g_{CW} and every nonlinear combiner. Let $A^* = \sup_g A(g)$ be the Neyman–Pearson ceiling, attained by the posterior ranker $g^*(s) = \Pr[Y=1 \mid s]$; no measurable rule exceeds A^* .

3 The Impossibility: Clean Transfer

Decompose the headroom of CW to the perfect ranker:

$$1 - A(g_{\text{CW}}) = \underbrace{(1 - A^*)}_{\varepsilon_{\text{Bayes}}} + \underbrace{(A^* - A(g_{\text{CW}}))}_{\varepsilon_{\text{subopt}} \geq 0}. \quad (1)$$

Because $A(g) \leq A^*$ for all g , the advantage of *any* class $\mathcal{G}$ over CW is capped by the optimality gap, not by the gap to perfect:

$$\Delta^*(\mathcal{G}) = \sup_{g \in \mathcal{G}} A(g) - A(g_{\text{CW}}) \leq \varepsilon_{\text{subopt}} \leq 1 - A(g_{\text{CW}}). \quad (2)$$

Theorem 1 (Clean-transfer impossibility). *Under the Gaussian equal-covariance model $(S \mid Y=y) \sim \mathcal{N}(\mu_y, \Sigma)$, the posterior ranker is linear and $A^* = \Phi(\frac{1}{\sqrt{2}} \|\mu_1 - \mu_0\|_{\Sigma^{-1}})$. If the confidence weights align with $\Sigma^{-1}(\mu_1 - \mu_0)$ (the redundant, equal-discriminability case) then $A(g_{\text{CW}}) = A^*$ exactly and $\Delta^*(\mathcal{G}) \leq 0$ for every $\mathcal{G}$. More generally, a one-sided level- α , power- β paired AUROC test certifies $\Delta > 0$ only if $\varepsilon_{\text{subopt}} > \text{MDE}(\alpha, \beta, n, \rho)$; hence clean Gate “superiority” is unattainable whenever $\varepsilon_{\text{subopt}} < \text{MDE}$.*

A guarded, computable certificate. $\varepsilon_{\text{subopt}}$ is estimable: train an *unconstrained* oracle (gradient boosting on the joint scores and confidences with labels, cross-fitted) as an over-generous estimate of A^* . If even the oracle cannot beat CW, no restricted gate can. Crucially, the certificate is only valid when the oracle is a credible ceiling: because the oracle has strictly more information than CW, a credible oracle must match CW up to finite-sample noise. We therefore declare the certificate *inconclusive* (rather than “unpassable”) whenever the oracle is detectably weaker than CW ($A_{\text{CW}} - A_{\text{oracle}} > \text{MDE}$). This guard makes the impossibility *non-manufacturable* by under-powering the oracle—a soundness property absent from a naive headroom test.

4 The Matching Positive: Stress

The complement of Theorem 1 is that reliability blindness is provably costly exactly when modalities degrade.

Theorem 2 (Reliability-blind impossibility under corruption). (a, sparse). *For the 1-of- D collapse adversary (one modality becomes $\text{Uniform}[0, 1]$ noise with probability p), every linear reliability-blind aggregator has L_2 risk gap at least $p/(3D^2)$ over a reliability-aware oracle, achieved with equality by the mean.* (b, coherent). *For the coherent adversary (with probability p , all D modalities collapse to a single shared $\text{Uniform}[0, 1]$ draw), every reliability-blind aggregator—including the median and any rank rule—has risk gap at least $p/12$. The oracle abstains because every modality reads reliability 0.*

A finite-sample switching certificate (empirical-Bernstein) and a closed-form stochastic dilution law $\Pr[\text{gate silent} \mid k] = \Phi((\mu_k - \tau)/(\sigma/\sqrt{D}))$ quantify, respectively, when a deployment may safely switch and how many modalities must degrade before a mean-gate reacts.

5 The Operating Boundary

Theorems 1 and 2 meet:

Corollary 1 (Operating boundary). *Reliability gating’s certifiable advantage over the confidence-weighted mean is zero on clean near-ceiling transfer (Theorem 1) and lower-bounded positive under coherent or k -of- D stress (Theorem 2). Reliability gating is therefore a stress-regime mechanism with a sharp, provable operating boundary, not a universal fusion improvement.*

This is a statement about the *problem*, independent of any particular gate architecture: it tells a practitioner not to expect clean-transfer gains and tells a theorist exactly where the mechanism’s value lives.

6 A Degenerate-Channel Robustness Guard

A second, orthogonal failure mode corrupts *any* aggregator: a single expert whose scores are sign-inverted (ranks anomalies below normals) or saturated (near-constant). The confidence-weighted mean trusts such a “confidently wrong” expert and can collapse below chance. We give a validation-only guard that diagnoses each channel (inversion via $\text{AUROC} < \frac{1}{2}$ on validation; saturation via dispersion and boundary-pile statistics) and zeroes its weight, using no test labels. The guard composes with any upstream expert and makes even the plain confidence-weighted mean robust; it is the floor on which any fusion rule should sit.

7 Empirical Illustration

We illustrate—rather than establish—the boundary on industrial multimodal anomaly benchmarks (RGB+depth, RGB+infrared, and RGB+photometric-stereo+ point-cloud). The pattern matches the theory: against the strongest validation- frozen baseline in-domain a reliability gate improves significantly; under induced or natural differential degradation it improves over the mean; and on clean cross-domain transfer it provably ties (the oracle-guarded certificate returns *unpassable* with the estimated optimality gap below the minimum detectable effect on every clean cell). The degenerate-channel guard removes an inversion artifact that would otherwise inflate any pooled comparison. We report all negatives; the empirical role here is corroboration of the boundary, not a superiority claim.

8 Related Work and Scope

Impossibility and limit results—no-free-lunch theorems, the robust-statistics breakdown literature, and selective-prediction bounds—inform the framing: a characterized boundary is a stronger, more durable contribution than a benchmark win. Our scope is the ranking (AUROC) objective and the reliability-gating family; we do not claim a universal detector, and the clean- transfer impossibility is, by construction, a limit rather than a method.

9 Conclusion

Reliability-aware multimodal fusion has a provable operating boundary: it cannot beat the confidence-weighted mean on clean near-ceiling transfer, and it provably does beat reliability-blind aggregation under differential-reliability stress. The contribution is a law of the problem plus a computable, soundness-guarded certificate and a reusable robustness guard—results that hold for the whole fusion family, not a single tuned method.